

Numerical Linear Algebra First-Year Exam, January 2018

Do 4 (four) problems.

1. Assume $A \in \mathbb{R}^{m,n}$ with $m \geq n$, $\text{rank}(A) = n$, and $b \in \mathbb{R}^m$.
 - (a) Define the least squares solution to $Ax = b$.
 - (b) Derive the normal equations for the least squares problem.
 - (c) Prove that $A^T A$ is invertible.
 - (d) Prove that the unique solution to the least squares problem is $(A^T A)^{-1} A^T b$.
 - (e) Describe how to solve the least squares problem using the QR decomposition of A .
2. Define a normal matrix and prove that the following are equivalent.
 - (a) A is normal.
 - (b) $\|Ax\|_2 = \|A^*x\|_2$ for every x .
 - (c) A is unitarily diagonalizable.
3. Assume $A \in \mathbb{R}^{m,m}$.
 - (a) Prove that $\langle x, y \rangle_A = x^* A y$ is an inner product on $\mathbb{R}^m$ if and only if A is symmetric and positive definite.
 - (b) Assume now that A is symmetric and positive definite. If x_* is the solution to $Ax = b$ and $\{p_1, \dots, p_m\}$ is an orthonormal basis for $\mathbb{R}^m$ with respect to $\langle \cdot, \cdot \rangle_A$ and $x_* = \sum c_i p_i$, give a formula for the c_i .
4. If $q_1, \dots, q_n$ is an orthonormal basis for the subspace $V \subset \mathbb{C}^m$ with $m > n$, prove that the orthogonal projector onto V is QQ^* , where Q is the matrix whose columns are the q_j .
5. Assume $A \in \mathbb{C}^{m,m}$.
 - (a) Show that A has a Schur decomposition.
 - (b) If A has a collection of m linearly independent eigenvectors, show that A is diagonalizable.